

YUVARAJ R

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SUMMARY

Computer Science and Engineering student at VIT Chennai specializing in AI-driven systems, backend development, and edge computing. Experienced in building end-to-end applications combining machine learning, scalable backend architectures, and embedded systems. Strong Linux user with experience in self-hosted infrastructure, computer vision, and distributed applications.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++, SQL, Dart, Solidity
AI & ML: YOLOv8, Scikit-learn, ANN, Computer Vision, NLP, Whisper, Groq (Llama 3)
Frontend: Next.js, React, React Native (Expo), Flutter, Tailwind CSS, Three.js (R3F)
Backend: FastAPI, Flask, Node.js, Express, PostgreSQL, PostGIS, Neo4j, Redis, Celery
DevOps & Systems: Docker, Linux (NixOS), Self-hosted services
Embedded: Raspberry Pi, ESP8266, Arduino, Basic PCB Design (KiCad)
Blockchain: Solidity, Polygon (Amoy Testnet), Smart Contracts, IPFS

PROJECTS

MechSight AI | *Python, YOLOv8, Flask, Raspberry Pi*

- Built an end-to-end industrial quality inspection system using YOLOv8 for defect detection in automotive components.
- Developed centralized AI inference server with Flask APIs and automated defect logging.
- Designed Raspberry Pi edge client for physical control and real-time visual feedback.

Wayline | *OSRM, PostgreSQL, Docker*

- Built a self-hosted routing and geocoding platform using OSRM and PostgreSQL.
- Implemented API gateway architecture for routing services with scalable container deployment.

BhoomiNode | *FastAPI, React Native, Groq (Llama 3), PostGIS, Polygon, Docker* AGRHINOVA Hackathon, Apr 2026

- Built a blockchain AgriTech platform enabling farmers to register land via voice, obtain an AI-scored Kisan Passport, and file verified crop insurance claims.
- Integrated Groq Whisper for regional-language speech-to-text and Llama 3 for intent extraction, with NDVI scoring via Microsoft Planetary Computer satellite data.
- Deployed on-chain land NFTs and insurance verdicts via Solidity smart contracts on Polygon Amoy Testnet with a PostGIS and Neo4j backend.

Hunter System | *React 19, Node.js, Express, TypeScript, PostgreSQL, Docker*

- Built a full-stack life RPG gamification platform that converts real-life tasks and goals into a quest system with XP, leveling, and stat tracking.
- Designed a 12-table PostgreSQL schema covering quests, boss raids, player stats, inventory, and skill trees with JWT authentication.
- Containerized the full stack using Docker Compose for single-command deployment.

Atlas Finance | *Next.js, FastAPI, Flutter, PostgreSQL, Ollama, Docker*

- Architected a dual-frontend financial tracking platform using Next.js and Flutter.
- Integrated local LLMs via Ollama for on-device financial analysis without cloud data exposure.
- Designed FastAPI backend with PostgreSQL ledger and Docker-based deployment.

Vidora AI | *Next.js, FastAPI, Python, Docker*

- Built an intelligent video analysis platform using a hybrid local and cloud AI strategy.
- Implemented a full-stack monorepo with a Next.js frontend and FastAPI backend, containerized with Docker Compose.

NoteWeave API | *FastAPI, Whisper, NLP, Celery, Redis, Docker*

- Built asynchronous API pipeline for lecture processing using Whisper and NLP extraction.
- Designed scalable microservice architecture using Celery, Redis, and Docker.

Miltz | *Next.js 14, React Three Fiber, Framer Motion*

- Developed high-performance marketing website with immersive 3D visuals using React Three Fiber.
- Implemented advanced scroll-linked animations and optimized rendering performance.

RESEARCH & SIMULATION

Electrical Fault Detection & Classification

- Modeled power systems in MATLAB Simulink to generate fault datasets.
- Compared eight ML models; Decision Trees and Polynomial Regression showed best performance.

CGM Sensor with ML Drift Correction

- Researched machine learning approaches for correcting sensor drift in continuous glucose monitoring systems.

EDUCATION

Vellore Institute of Technology, Chennai
B.Tech Computer Science and Engineering

2024 – 2028
Chennai, India

ADDITIONAL EXPERIENCE

- Rajya Puraskar Awardee — Bharat Scouts & Guides
- Participant, AGRHINOVA Hackathon — CeAT-RC, VIT Chennai (Apr 2026)
- Participant, Stock Market Simulation Finance Fest — VNEST, VIT Chennai (Dec 2025)
- Participant, Expert Talk Quality Week 2026 — Bureau of Indian Standards, SCOPE, VIT Chennai (Mar 2026)
- Advanced Linux user; daily development environment based on NixOS